

HydroPowerDynamics.jl Francis model

$$H = (p_{\text{in}} - p_{\text{out}})/(\rho g)$$

$$Q = u K_q D^2 \sqrt{|H| + \epsilon}$$

$$\dot{m} = \rho Q$$

$$\eta = \eta_{\text{max}}[1 - c_\eta(Q/Q_r - 1)^2]$$

$$P_m = \rho g Q |H| \eta$$

$$\tau = P_m / (|\omega| + \epsilon)$$

physics to implementation

MTK ports and equations

```
port_in, port_out = HydraulicPort()
```

```
opening = SignalInPort()
```

```
shaft = RotationalPort()
```

```
port_in.dm + port_out.dm ~ 0
```

```
port_in.dm ~ dm
```

```
shaft.tau ~ -tau_shaft
```

RotorInertia

states: θ, ω

$$\dot{\theta} = \omega$$

$$J\dot{\omega} = \tau_t + \tau_g - b_v\omega$$

$u = \tau_o$

SimpleGovernorAGC

$$e_f = (\omega_{\text{ref}} - \omega)/\omega_{\text{ref}}$$

$$\dot{\xi} = e_f$$

$$u_c = u_b + e_f/R + K_i\xi$$

$$u_s = \text{clamp}(u_c, u_{\text{min}}, u_{\text{max}})$$

$$T_g\dot{u} = u_s - u$$